A Project Report

on

Exam Cell Automation System

By

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DEPARTMENT OF INFORMATION TECHNOLOGY

THAKUR COLLEGE OF SCIENCE AND COMMERCE

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CERTIFICATE

This is to certify that the project entitled, **"Exam Cell Automation System",** undertaken at the Thakur College of Science and Commerce by **Viranshi Sanghavi** Roll. No: **4414** is submitted in partial fulfilment of the requirements for the award of degree of BACHELOR OF SCIENCE in INFORMATION TECHNOLOGY SEM VI Examination and does not form part of any other course undergone by the candidate.

It is further certified that he has completed all the required phases of the project.

Project Guide **HOD**

External Examiner **Internal Examiner**

College Seal

ABSTRACT

The academic management landscape is rapidly evolving, with increasing emphasis on digital platforms. Our new app, "EXAM TIME," represents a significant advancement in educational technology. It focuses on streamlining the entire exam management process, including backlog management, report generation, and secure question paper handling.

With our app, faculty and administrators can efficiently manage the exam process, reducing the complexity and manual effort involved in tasks such as question paper uploads and backlog tracking. The system also provides comprehensive reports, offering clear insights into student performance and administrative efficiency. Our aim is to make the exam process more organized and efficient, catering specially to the needs of academic institutions.

"EXAM TIME" is not just another educational tool – it’s a targeted solution designed to transform the way exams are managed, making the process more streamlined, secure, and effective, setting a new standard in academic administration.

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Lastly, we extend our thanks to our friends for their constant support, motivation, and encouragement, which helped us stay focused and determined to complete this project successfully

DECLARATION

We hereby declare that the project entitled, "**EXAM CELL AUTOMATION SYSTEM**" is done at **Thakur College of Science and Commerce**, this written submission represents our ideas in our own words and where other's ideasor words have been included, we have properly credited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/data/fact/source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evokepenal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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Signature

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SYNOPSIS

Introduction: The Exam Cell Automation System is designed to streamline and enhance the entire examination workflow, reducing manual efforts and minimizing errors. From setting up and distributing question papers to tracking deadlines and generating reports, this system transforms exam management by automating tasks, ensuring accuracy, and saving time for faculty and administrators. With this system, institutions can focus more on delivering quality education while the complexities of exam administration are handled seamlessly.

Overview: The Exam Cell Automation System operates continuously, efficiently managing all aspects of the examination process, something that would be challenging to handle manually. A key feature of this system is its ability to reduce human error and bias. Unlike manual processes, which can be prone to mistakes or inconsistencies, this system is driven purely by data and predefined work patterns. This ensures a high level of accuracy and consistency in exam administration, from scheduling to result generation, ultimately leading to a smoother and more reliable examination process.

Objectives: The primary objective of an Exam Cell Automation System is to enhance the efficiency of examination-related processes within an educational institution.

Design and Architecture: We are designing the Exam Cell Automation System by focusing on two key principles: Flexibility and Incremental Development .These principles are based on Agile Model which is crucial for adapting to the dynamic needs of an educational institution. The architecture of the Exam Cell Automation System features a modular design with a front-end UI for user interaction, a business logic layer for processing exam-related functions, a data layer for managing and storing information, and a security layer to protect data and control access.

Code and Functionality: Our Exam Cell Automation System is built using Java, Firebase, and MySQL. Users register and log in, manage exam schedules, and access various features through our application, with data stored securely in Firebase and MySQL.

Future Work: In the future, we plan to add more features to the Exam Cell Automation System, optimize its existing functionalities, and enhance its performance. This will further improve the system's capabilities and user experience on mobile platforms.

Conclusion: Our Exam Cell Automation System provides a seamless and efficient solution for managing examination processes. It represents a significant advancement in automating exam management, potentially leading to more streamlined and accurate operations. With ongoing development and enhancements, our system has the potential to become a valuable tool for educational institutions, making exam management more efficient and effective. It is the future of examination management, today.

CHAPTER 1

INTRODUCTION

In the realm of educational institutions, managing examination processes efficiently is crucial for maintaining academic standards and ensuring a smooth operational flow. An Exam Cell Automation System is designed to streamline and enhance the administrative tasks associated with the examination cell. By leveraging advanced technology, this system automates various aspects of the examination process, reducing the manual workload and minimizing errors.

The primary objective of an Exam Cell Automation System is to provide a comprehensive solution for managing exam-related activities, including scheduling, invigilation, result processing, and student communication. This system integrates various functionalities to support the seamless execution of exams, from initial planning to the final result announcement. It enables administrators to create and manage exam schedules, allocate exam rooms, and monitor the examination process in real time.

Another significant benefit of the Exam Cell Automation System is its ability to handle large volumes of data efficiently. It ensures accurate and timely processing of examination results, which is essential for maintaining transparency and fairness in the evaluation process. The system can generate detailed reports and analytics, providing insights into student performance and helping educators make informed decisions.

Moreover, the automation system enhances the user experience for both students and staff. Students benefit from features such as online access to examination schedules, and the ability to receive instant notifications about exam-related updates. For staff, the system simplifies administrative tasks, reduces paperwork, and facilitates better coordination between different departments involved in the examination process.

In contemporary educational environments, the efficiency and accuracy of examination processes are critical to academic success and institutional credibility. An Exam Cell Automation System represents a significant advancement in the way educational institutions manage their examination activities. This system leverages modern technology to automate and streamline various functions related to exams, thus transforming traditional, manual methods into a more efficient, error-resistant, and transparent process.

In summary, an Exam Cell Automation System is a valuable tool for modern educational institutions aiming to optimize their examination procedures. By automating repetitive tasks and providing real- time data management, the system improves efficiency, accuracy, and overall operational effectiveness, ultimately contributing to a more streamlined and organized examination process.

1.1 OBJECTIVE & SCOPE & GOAL:

♣ **Objective:**

1. To Revolutionize Examination Management with Automation

- To revolutionize the examination management process by integrating automation into key administrative tasks.

2. To Streamline Exam Scheduling, Resource Management, and Result Processing

- To streamline the scheduling of exams, management of resources, and processing of results for enhanced efficiency and accuracy.

3. To Reduce Manual Workload and Minimize Human Errors

- To reduce the manual workload by automating routine tasks, minimizing human errors, and ensuring a more organized examination environment.

4. To Enhance Accuracy and Efficiency in Examination Operations

- To improve the accuracy and efficiency of examination operations by providing precise tools for scheduling, resource allocation, and result processing.

5. To Ensure Fair Conduct and Timely Delivery of Results

- To ensure fair conduct of exams and prompt delivery of results, contributing to a more reliable examination process.

6. To Improve User Experience for Students

- To enhance the user experience for students by providing easy access to exam schedules, timely notifications, and a transparent results process.

- **Scope:**

The scope of the Exam Cell Automation System includes automating various aspects of examination management to enhance efficiency and accuracy. This involves automating exam scheduling to minimize conflicts ensuring exams are well-organized and resources are effectively utilized. The system also encompasses secure management of exam papers, including their creation, distribution, and storage, protecting the integrity of the exam content throughout the process.

Additionally, the system covers automation of result processing and grade management. It enables quick and accurate processing of exam results, reducing manual errors, and automates the calculation, tracking, and distribution of grades.

Furthermore, the system includes dedicated portals for students and faculty to streamline their interactions with exam-related information. The Student Portal offers access to exam schedules, results, and study materials, enhancing student preparation and engagement. The Faculty Portal allows educators to manage exam schedules, student registrations, and grades efficiently. The system also provides comprehensive reporting and analytics to generate insights into exam performance and trends, alongside robust security and access control measures to protect sensitive data and ensure authorized access.

- **Goal:**

The main goal of the Exam Cell Automation System is to help administrators manage exams efficiently and effectively. By providing access to automated tools and real-time insights, the system aims to enable administrators to make informed decisions, reduce errors, and ensure a seamless exam experience.

♣ **Specifically, the app wants to:**

1) Help administrators identify potential exam irregularities or discrepancies, enabling prompt action.
2) Assist administrators in managing exam-related risks by setting up automated checks and balances.
3) Send out alerts to administrators when important exam-related events occur, such as exam schedule changes or student registration deadlines.
4) Allow administrators to customize the system to fit their specific exam management needs and workflows.
5) Provide administrators with insights and analytics to evaluate exam performance, identify areas for improvement, and refine their exam management strategies.
6) By doing these things, the Exam Cell Automation System hopes to help administrators streamline exam management, reduce errors, and improve the overall exam experience.

1.2 THEORETICAL BACKGROUND:

The Exam Cell in educational institutions is a critical administrative unit responsible for the management and execution of all examination-related activities. The traditional approach to exam cell operations involves significant manual effort, which includes scheduling exams, coordinating with faculty for question paper setting, ensuring proper invigilation, collecting and grading answer sheets, and finally, processing and publishing results. This manual process is often labor-intensive and susceptible to human errors, which can lead to issues such as mismanagement of schedules, delayed result processing, and inaccuracies in grade calculation. The theoretical foundation for automating these processes lies in the principles of operational efficiency, data accuracy, and process standardization, which are essential for maintaining the integrity of academic assessments.

One of the key theoretical frameworks that underpin the concept of an Exam Cell is Operational Efficiency. This concept emphasizes the importance of streamlining processes to maximize productivity while minimizing resource consumption, time, and effort. In the context of exam management, operational efficiency can be achieved by automating routine tasks, reducing manual data entry, and enabling real-time communication between different stakeholders. The theory suggests that by optimizing the use of technology, educational institutions can handle the growing complexity and scale of exam operations more effectively, thus improving overall institutional performance.

Another relevant theoretical concept is Information Systems and Data Management, which focuses on the use of technology to collect, store, manage, and retrieve large volumes of data. Exam cells handle a significant amount of data, including student information, exam schedules, question papers, and results. Theoretical models in data management, such as database management systems (DBMS) and data security protocols, provide the framework for creating systems that can efficiently manage this data while ensuring its integrity and confidentiality. Effective data management is crucial for accurate record-keeping, quick access to information, and secure storage of sensitive data, which are all essential aspects of an exam cell's operations.

The theory of Process Automation is also central to understanding the need for an automated Exam Cell system. Process automation involves the use of technology to perform tasks with minimal human intervention, ensuring consistency, accuracy, and speed. In an exam cell, automation can be applied to various processes, such as exam scheduling, result processing, and communication with students and faculty. Theoretical models of automation highlight the benefits of reducing human error, increasing efficiency, and allowing staff to focus on more strategic tasks, thereby enhancing the overall effectiveness of exam administration.

Change Management Theory is another important framework relevant to the implementation of an Exam Cell Automation System. This theory deals with the strategies and processes necessary for managing change within an organization, especially when introducing new technologies. Implementing an automated system in an exam cell requires a shift from traditional methods to more technology-driven approaches. Change management theory suggests that successful implementation depends on effective communication, training, and stakeholder involvement to overcome resistance and ensure smooth adoption. Understanding this theory is crucial for educational institutions looking to transition to automated systems without disrupting existing operations.

The Theory of Educational Assessment also provides a theoretical basis for the functions of an Exam Cell. This theory explores the principles and practices of evaluating student performance through exams and assessments. It emphasizes the importance of fairness, transparency, and accuracy in the assessment process, which are key considerations for an exam cell. Automation can support these principles by standardizing processes, ensuring consistent application of grading criteria, and providing clear, timely communication of results. By aligning with the theory of educational assessment, an automated exam cell can contribute to upholding the integrity and quality of academic evaluations**.**

Finally, the theoretical background of Agile Methodology is relevant for understanding the development and implementation of an Exam Cell Automation System. Agile is a project management and software development approach that emphasizes iterative progress, flexibility, and continuous improvement. In the context of exam cell automation, Agile principles can be applied to develop the system in stages, allowing for ongoing feedback and adjustments based on the needs of users.

1.3 PROBLEM DEFINITION:

The Exam Cell Automation System is designed to address the inefficiencies and challenges associated with the manual management of examination processes in educational institutions. Traditional exam cell operations are often labor-intensive and prone to human errors, leading to delays, inaccuracies, and difficulties in handling large volumes of data. As institutions grow, these issues become increasingly problematic, affecting the overall quality and integrity of the examination process. By automating these processes, the Exam Cell Automation System aims to streamline operations, reduce errors, and improve the efficiency of exam management.

Key Problems in Manual Exam Cell Management:

1. **Inefficiency and Time Consumption:**

- Manual scheduling, invigilation management, and result processing are slow and labor-intensive.
- Coordinating with multiple stakeholders often leads to delays.

2. **Human Errors:**

- Errors in data entry, result calculation, and record-keeping are common and can lead to significant issues.
- Miscommunication between departments can result in scheduling conflicts and mistakes in exam logistics.

3. **Data Management Challenges:**

- Difficulty in storing, retrieving, and managing large volumes of student and exam-related data.
- Lack of integration with other academic systems creates data silos and inefficiencies.

4. **Security Concerns:**

- Manual handling of sensitive exam materials, such as question papers, increases the risk of leaks.

- Ensuring the confidentiality and integrity of exam data is challenging without proper security protocols.

5. **Student and Faculty Dissatisfaction:**

- Delays in publishing results and communicating exam schedules lead to frustration among students and faculty.

- Inefficient processes can negatively impact the overall academic experience.

♣ **Solution:**

1. Implement automated scheduling tools that allow for real-time updates and streamlined coordination with faculty and invigilators.

2. Utilize workflow automation for result processing, reducing the time required to compile and publish results.

3. Introduce validation rules and automated data entry processes to ensure accuracy in student records and result calculations.

4. Implement centralized communication platforms to reduce miscommunication and prevent scheduling conflicts.

5. Deploy a centralized database that integrates with other academic systems, allowing for seamless data storage, retrieval, and management.

6. Implement data analytics tools to help administrators easily manage large volumes of exam-related data and generate reports.

7. Use encryption and secure access protocols to protect sensitive exam materials, including question papers and student data.

8. Implement role-based access controls to ensure that only authorized personnel can view or modify exam-related information.

9. Provide a student portal where students can access their exam schedules and results in real time, reducing delays and frustration.

10. Offer a faculty portal for easy management of exam-related tasks, fostering better communication and coordination with the administration.

1.4 USER REQUIREMENT:

User requirements for the Exam Cell Automation System are derived from the specific needs and expectations of its users, including students, faculty, and administrative staff. These requirements, identified through user research and feedback, guide the design and development of the system to ensure it meets user goals effectively. For students, the system must provide easy access to exam schedules, online registration, real-time notifications, and result viewing. Faculty require functionalities for managing exam papers, processing grades, and performance analytics. Administrative staff need features for automated scheduling, student data management, security controls, result compilation, system integration, and audit tracking.

♣ **Functional User Requirement:**

1. **Automated Exam Scheduling:** The system must facilitate automated creation and management of exam schedules and timetables, ensuring no conflicts and efficient use of resources.
2. **Online Exam Registration and Result Viewing:** Students should be able to register for exams online and access their results in real-time through a dedicated portal.
3. **Secure Exam Paper Management:** Faculty need tools to securely create, upload, manage, and review exam papers, including setting questions and marking schemes.
4. **Grading and Result Processing:** The system must support accurate and efficient grading, including batch processing and result verification for faculty.
5. **System Integration:** Seamless integration with other academic systems such as Learning Management Systems (LMS) and Student Information Systems (SIS) for data consistency.
6. **Real-Time Notifications and Communication:** Automated notifications and a communication platform to keep students and faculty informed about exam schedules, changes, and results.
7. **Feedback Collection:** Capability for students to provide feedback on the exam process, invigilation, and content.

♣ **Non-Functional User Requirement:**

1. **Security and Data Protection:** The system must ensure high levels of security, including encryption and role-based access controls to protect sensitive exam materials and student data.
2. **User Interface and Usability:** The system should have an intuitive and user-friendly interface to facilitate ease of use for students, faculty, and administrative staff.
3. **Scalability:** The system must be scalable to accommodate increasing numbers of users and growing volumes of data without performance degradation.
4. **Performance and Reliability:** The system should offer reliable performance with minimal downtime and fast response times for all operations.
5. **Compliance and Standards:** The system must adhere to relevant educational standards and data privacy regulations to ensure compliance and protect user information.
6. **Data Backup and Recovery:** The system must have robust data backup and recovery procedures to prevent data loss and ensure business continuity in case of system failures.
7. **Performance Monitoring:** The system should include tools for monitoring performance metrics and system health to proactively address any issues that may affect its functionality.
8. **Support and Maintenance:** The system must provide reliable support and maintenance services, including regular updates and troubleshooting assistance to ensure ongoing functionality and user satisfaction.

1.5 FEASIBILITY STUDY:

A feasibility study for the Exam Cell Automation System aims to evaluate whether the project is viable, achievable, and sustainable. This comprehensive analysis will assess the technical, financial, legal, and operational aspects to determine if automating the exam management process is worthwhile.

1. **Market and Asset Selection:** Identify the specific educational institutions and their exam management needs that the system will target. Determine whether the system will cater to schools, colleges, or universities, and define the types of exams and academic assessments it will handle.

2. **Technical Feasibility**: Evaluate the technical requirements for developing the Exam Cell Automation System. This includes understanding the necessary programming languages (e.g., C#, SQL), development frameworks, and integration with existing academic systems such as Learning Management Systems (LMS) and Student Information Systems (SIS).

3. **Historical Data Analysis:** Review historical data on current exam management practices and their inefficiencies. Assess how existing manual processes impact administrative efficiency, data accuracy, and user satisfaction to gauge the potential benefits of automation.

4. **System Requirements:** Define the specific features and functionalities needed for the system, such as automated scheduling, secure exam paper management, result processing, and user portals for students and faculty. Establish clear requirements and parameters for each feature.

5. **Prototype Testing:** Develop and test a prototype of the Exam Cell Automation System to evaluate its performance and functionality. Perform usability testing with a sample group of users to ensure that the system meets their needs and operates as expected.

6. **Cost Projections:** Estimate the financial implications of developing and deploying the system. This includes costs for software development, hardware infrastructure, maintenance, and potential training for users. Compare these costs with the anticipated savings and efficiency gains from automation.

7. **Risk Assessment:** Analyze the risks associated with implementing the Exam Cell Automation System. Consider technical risks, such as integration challenges and system reliability, as well as operational risks, including user adoption and data security. Develop a risk management plan to address these concerns.

8. **Feasibility Conclusion:** Based on the analysis of technical requirements, historical data, system requirements, prototype testing, cost projections, and risk assessment, provide a conclusion on the feasibility of proceeding with the development and implementation of the Exam Cell Automation System. This conclusion should determine whether the project is viable and worth pursuing.

1.6 DETAILS OF HARDWARE AND SOFTWARE:

The hardware and software requirements for an Exam Cell Automation System can vary depending on the specific needs of the system and the educational institution it is designed to support. However, there are some general hardware and software requirements to consider.

♣ **Hardware requirements (minimum) for making Exam Cell Automation App:**

1. Processor – i5 12450H
2. Hard Disk – 475 GB
3. RAM- 16 GB
4. SSD ROM- 512 GB

♣ **Software versions required (minimum) for making Exam Cell Automation App:**

1. Java
2. Android Studio
3. Firebase
4. Windows 10 Operating system
5. MySQL Database Setup

CHAPTER 2

SYSTEM ANALYSIS AND DESIGN

What is a System?

A system is an organized collection of interrelated components or elements that work together to achieve a specific purpose or goal. These components can be physical objects, processes, or even abstract concepts, and they interact in a structured manner according to a set of rules or a plan. System analysis and design for an Exam Cell Automation System is a critical process that ensures the effective development and deployment of an automated system that streamlines exam-related tasks.

♣ System Analysis:

1. **Understanding Requirements, Constraints, and Objectives.**

- Identify User Needs: Gather feedback from key stakeholders, including exam cell administrators, faculty, and students, through surveys or interviews. Understand their needs, challenges, and expectations regarding exam management.

- Define Project Scope: Clearly outline the functionalities the Exam Cell Automation System should include, such as exam scheduling, result processing, and secure exam paper management.

- Consider Constraints: Identify constraints such as budget, time limitations, and available resources. Ensure that the project's goals are realistic and achievable within these limitations.

2. **Analyzing Current Exam Management Practices.**

- Evaluate Existing Processes: Analyze current exam management practices to understand the challenges and inefficiencies in scheduling, paper handling, result processing, and communication.

- Identify Improvement Opportunities: Look for areas where automation can reduce errors, save time, and improve overall efficiency. Use this analysis to guide the design of the new system.

3. Identifying Technical Infrastructure Requirements.

- Determine Data Sources: Identify the sources of data required for the system, such as student information, course schedules, and grading data. Assess how this data will be integrated into the system.

- Assess Integration Needs: Determine the need for integration with other academic or administrative systems, such as the student information system or attendance management system.

- Implement Security Measures: Establish robust security protocols to protect sensitive data, including exam papers, student records, and grading information. Ensure compliance with relevant data protection regulations.

4. **User Interface Design and Experience.**

- Design User-Friendly Interfaces: Create intuitive interfaces for administrators, faculty, and students, making it easy to schedule exams, manage papers, and access results.

- Enhance User Experience: Focus on clear navigation and the easy retrieval of information. Use visual elements like calendars, dashboards, and notification systems to simplify complex processes.

- Incorporate Notifications: Implement alert systems to notify users about important events, such as exam dates, result announcements, and deadlines for paper submissions.

5. **Testing and Validation Strategies.**

- Conduct Comprehensive Testing: Test the system under various scenarios to ensure reliable performance, including stress testing for peak exam periods.

- Validate System Functions: Validate that each function, from scheduling to result processing, works as intended. This can include simulations of the entire exam cycle.

- User Acceptance Testing (UAT): Involve real users, such as administrators and faculty, in testing to gather feedback on usability, functionality, and overall satisfaction with the system. Make adjustments based on their feedback before full deployment.

♣ System Design:

1. Creating Smart Algorithms

We'll develop smart algorithms to automate exam scheduling by taking into account various factors like course timetables, faculty availability, and room resources. These algorithms will be designed to minimize scheduling conflicts and ensure efficient use of resources, helping administrators manage exams effectively.

2. Managing Data Flow

We'll organize how data, such as student information, course schedules, and exam details, moves within the system to ensure it's up-to-date and accurate. The system will also integrate with other academic platforms, such as student information systems, to provide real-time updates and seamless data exchange. This ensures that all stakeholders have access to the most current information.

3. User-Friendly Interface

Our system will feature an easy-to-use interface where users, including administrators, faculty, and students, can perform tasks like scheduling exams, uploading papers, and checking results effortlessly. The interface will include features like user registration, login, password recovery, and specialized pages for managing exam schedules, paper submissions, and result processing. The goal is to provide a smooth and intuitive experience for all users.

4. Keeping Things Secure

Security is crucial, so we'll implement measures such as strong passwords, data encryption, and role-based access controls to protect users' sensitive information, including student records, exam papers, and results. These security protocols will ensure that only authorized users can access specific functionalities and data, maintaining the integrity and confidentiality of the system.

5. Making Smart Calculations

Our system will perform automated calculations to process exam results accurately and efficiently. It will apply grading algorithms and handle complex calculations, such as grade point averages (GPA).

- Grade Calculation: The system will automatically calculate grades ensuring accuracy and consistency across all exam results.

- GPA and Ranking: By applying specific algorithms, the system can generate GPA and student rankings, giving administrators and students clear insights into academic performance.

6. Managing Exam Processes.

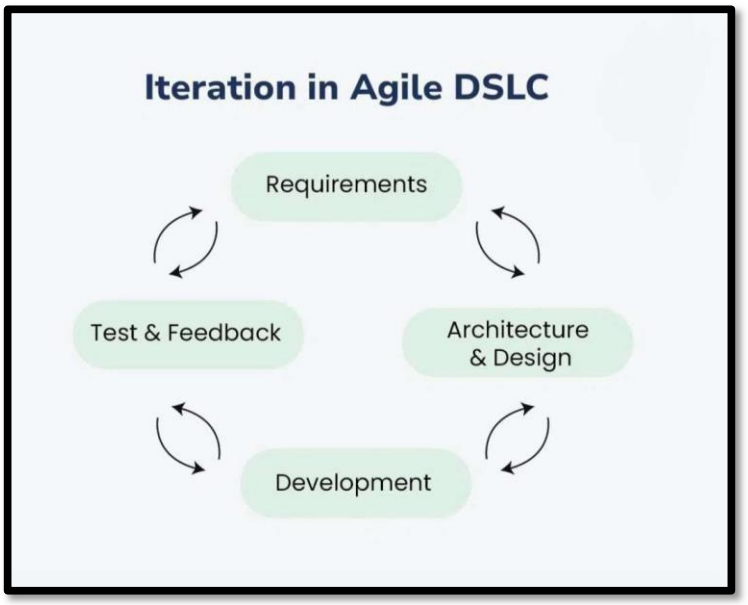
The system will provide administrators with intuitive tools to manage the entire exam process, from scheduling to result publication. Depending on the specific requirements, it will generate reports and insights, such as exam schedules, attendance records, and performance summaries. These tools will help administrators make informed decisions and manage exams more effectively.

7. Connecting with Academic Systems

We'll build seamless connections between our system and other academic platforms. Users will receive notifications and alerts to keep them informed about important exam-related activities. Once verified, data will be synchronized in real-time, ensuring that all information is current and accurate across all connected systems.

2.1 LIFE CYCLE OF PROJECT:

The development and operation of an Exam Cell Automation typically follow a well-defined life cycle. We are using the Agile SDLC model for our project. The Agile Software Development Life Cycle (SDLC) is an iterative and incremental software development methodology that prioritizes flexibility, collaboration, and customer feedback. Unlike traditional SDLC models, such as the waterfall model, which completes each step sequentially, the agile SDLC divides the development process into smaller iterations or increments.



Steps of Agile SDLC Model.

The agile model is a combination of iterative and incremental process models. The steps involve in agile SDLC models are:

- Requirement gathering
- Design the Requirements
- Coding
- Testing
- Deployment
- Feedback

Now, let's examine each step of the Agile Software Development Life Cycle (SDLC) in detail to understand the process thoroughly:

1. Requirement Gathering:

- In requirement Gathering collaboration takes place with stakeholders to understand and prioritize needs for the project, that focus on delivering value.
- In this step techniques like user stories and workshops for effective communication.

1. **Design the Requirements:**

- In design the requirement step that gathered requirements is translated into actionable tasks, breaking them down into smaller, manageable chunks.
- In this step developers develop visual representations of the solution, such as wireframes or prototypes, to get early feedback and ensure alignment with stakeholder expectations.

2. **Coding:**

- During the coding step, the development team implements the software in short, iterative cycles known as sprints. Each sprint focuses on delivering small, functional increments of the product.
- Developers write code, continuously integrate it into the main codebase, and regularly commit changes to detect and address issues early.

3. **Testing / Quality Assurance:**

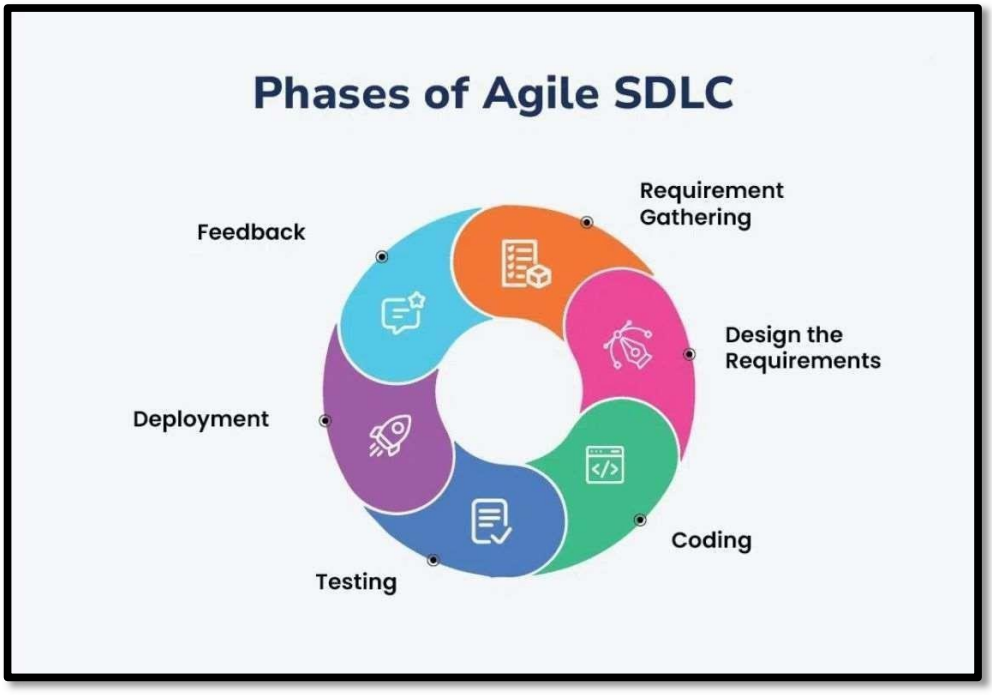
- Testing is an integral part of each iteration, ensuring the quality and functionality of the software.
- Unit tests are written to verify individual components, while integration tests check that different parts of the system work together as intended.
- User Acceptance Testing (UAT) involves end-users testing the software in real-world scenarios to ensure it meets their needs.

4. **Deployment:**

- Deployment involves releasing the software increments to production frequently and reliably.
- Automated deployment tools make the process, allowing for quick and consistent releases.
- Monitoring the deployment process and system performance helps identify and resolve any issues promptly.

5. **Feedback:**

- Feedback is crucial for continuous improvement in Agile.
- Stakeholders and end-users provide feedback through surveys, direct communication, and usage analytics.
- This feedback is used to refine requirements, prioritize changes, and identify areas for improvement.



Agile SDLC Process Flow

- **Concept:** In concept phase vision of the project, its scope, objectives, and goals are set.

- **Inception:** This stage is conducted before Construction. The major goal of this phase is to complete just enough work to get your team moving in the correct direction.

- **Iteration/Construction:** The main goal of this phase is to create a consumable solution with enough functionality to satisfy your stakeholders' present needs, or to configure it in the event of a package implementation.

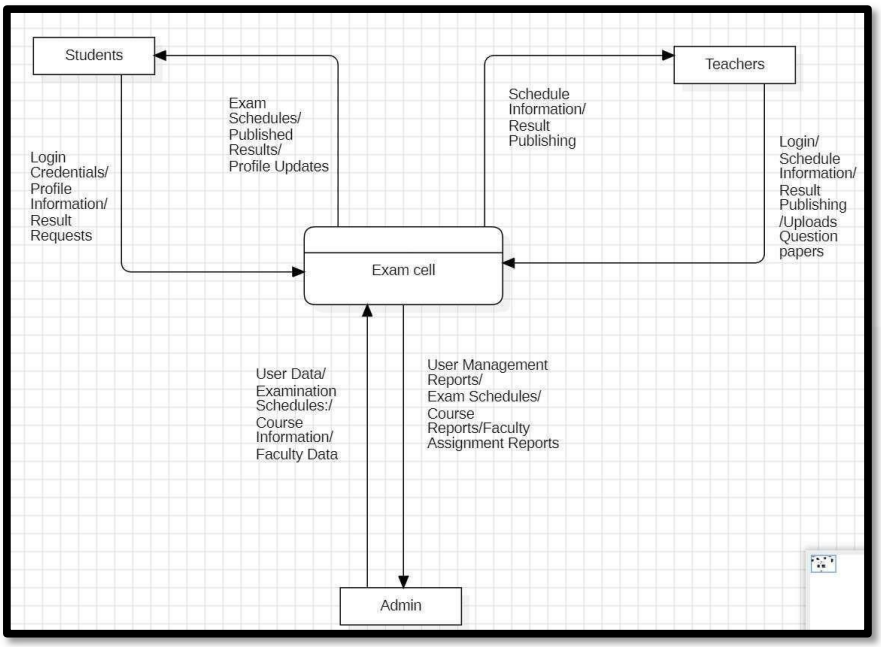
- **Release:** This stage normally involves final performance testing, security assessments, and documentation updates, as well as user acceptability testing (UAT), which ensures that user demands are satisfied and the product is successful. When the software is complete, it is released into production or made available to users.

- **Production:** The Production Phase aims to keep systems viable and productive after they have been deployed to the user population.

2.2 Dataflow Diagrams

A Data Flow Diagram (DFD) is a graphical representation that illustrates the flow of data within a system. In the context of the Exam Cell Automation System, a DFD can visually depict how data moves between different components or modules of the system, showing interactions between users, processes, and data stores.

LEVEL 0 DFD:



1. **External Entities-**

- Admin
- Teacher/Faculty
- Student

2. **Processes:**

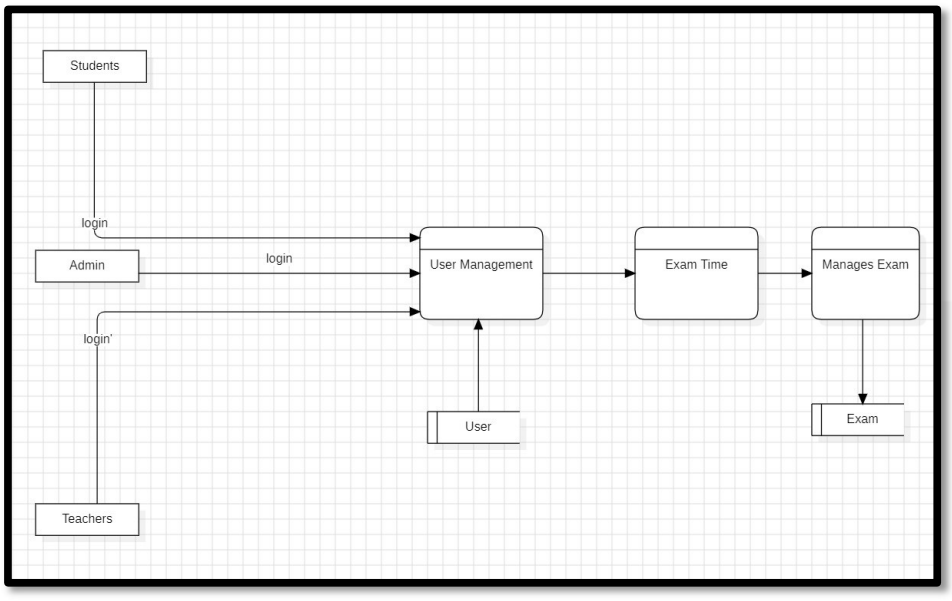
- Exam Cell

3. **Data Flows:**

- Admin to System:
    - User Data: Information about users (students, teachers) for management.
    - Examination Schedules: Details about exam schedules.
    - Course Information: Information about courses offered.
    - Faculty Data: Information about faculties.

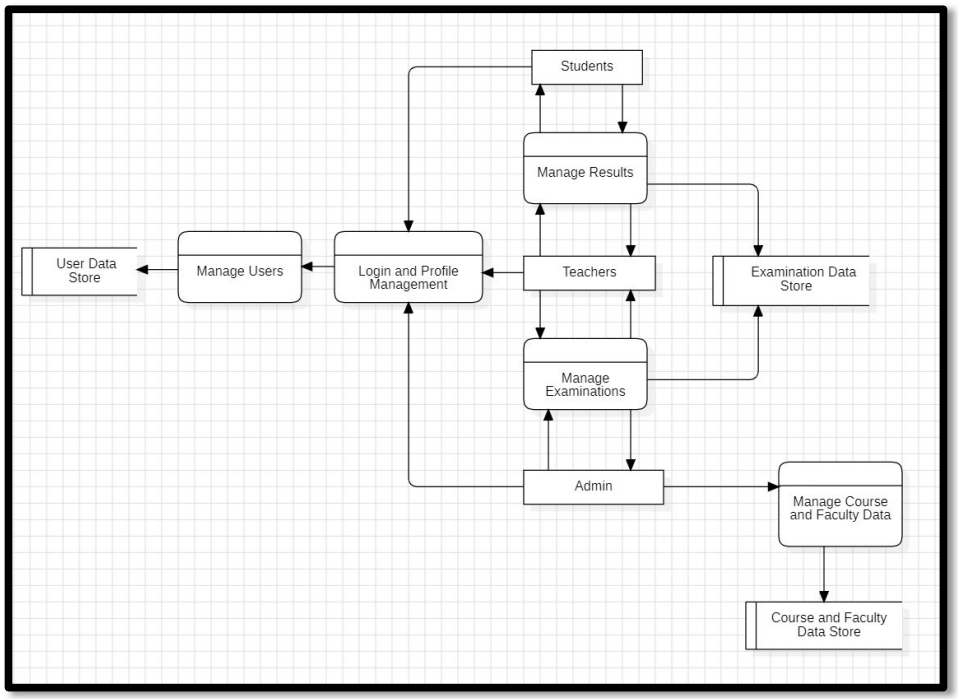
- System to Admin:
    - User Management Reports: Reports related to user data management.
    - Exam Schedules: Updated and managed exam schedules.
    - Course Reports: Details and updates about courses.
    - Faculty Assignment Reports: Information regarding faculty assignments.
- Teacher/Faculty to System:
    - Exam Papers: Uploads of examination papers.
    - Result Data: Marks and grades data for students.
    - Allotment Data: Data regarding the allotment of subjects or exams.
    - System to Teacher/Faculty:
    - Schedule Information: Exam schedules and allotment details.
    - Result Publishing: Confirmation and data regarding published results.
- Student to System:
    - Login Credentials: User authentication details.
    - Profile Information: Data related to student profiles.
    - Result Requests: Requests to check or download results.
- System to Student:
    - Exam Schedules: Information about upcoming exams.
    - Published Results: Access to results.

LEVEL 1 DFD:



1. **External Entities-**
   - Admin
   - Teacher/Faculty
   - Student
2. **Processes:**
   - User Management:
     - The admin manages user data by adding or updating users.
     - Connects with the Login and Profile Management.
   - Login and Profile Management:
     - Students and Teachers can manage their profile information.
     - Updates user data and ensures the information is sent back to the respective users.
   - Exam Time/Manage Examinations:
     - The Admin provides examination schedules and question papers.
   - Manages Exam:
     - Responsible for managing all exam-related activities.
3. **Data Stores:**
   - User Data Store:
     - Stores the credentials, profile information, and roles of all users.
   - Examination Data Store:
     - Holds examination schedules, question papers, and results.
   - Course and Faculty Data Store:
     - Store course information and faculty assignment data.
4. **Data Flows:**
   - From Students to Manage Results/Examinations:
     - Students request exam schedules and results.
   - From Teachers to Manage Examinations:
     - Teachers provide exam schedules, question papers, and results.
   - From Admin to User Management:
     - The admin manages users, their roles, and examination schedules.

LEVEL 2 DFD:



1. **External Entities:**

- Admin
- Teacher/Faculty
- Student

2. **Processes:**

- User Management
    - Receives user data (adding or updating users) from the Admin.
    - Sends updated user data to the User Data Store.
    - Connects with Login and Profile Management to provide user information.
- Login and Profile Management:
    - Fetches login credentials and profile information from Students and Teachers.
    - Sends updated profile data back to Students and Teachers.
- Manage Results:
    - Receives result data from Teachers and result requests from Students.
    - Publishes results to Students and updates the Examination Data Store.
    - Works closely with Teachers for result management and with Students for result access.

- Manage Examinations:
    - Receives examination schedules and question papers from Teachers and Admin.
    - Updates the Examination Data Store with schedules and papers.

3. Data Stores:

- User Data Store:
    - Stores user credentials, profile information, and roles.
- Examination Data Store:
    - Stores examination schedules, question papers, and results.
- Course and Faculty Data Store:
    - Stores course information and faculty assignment data.

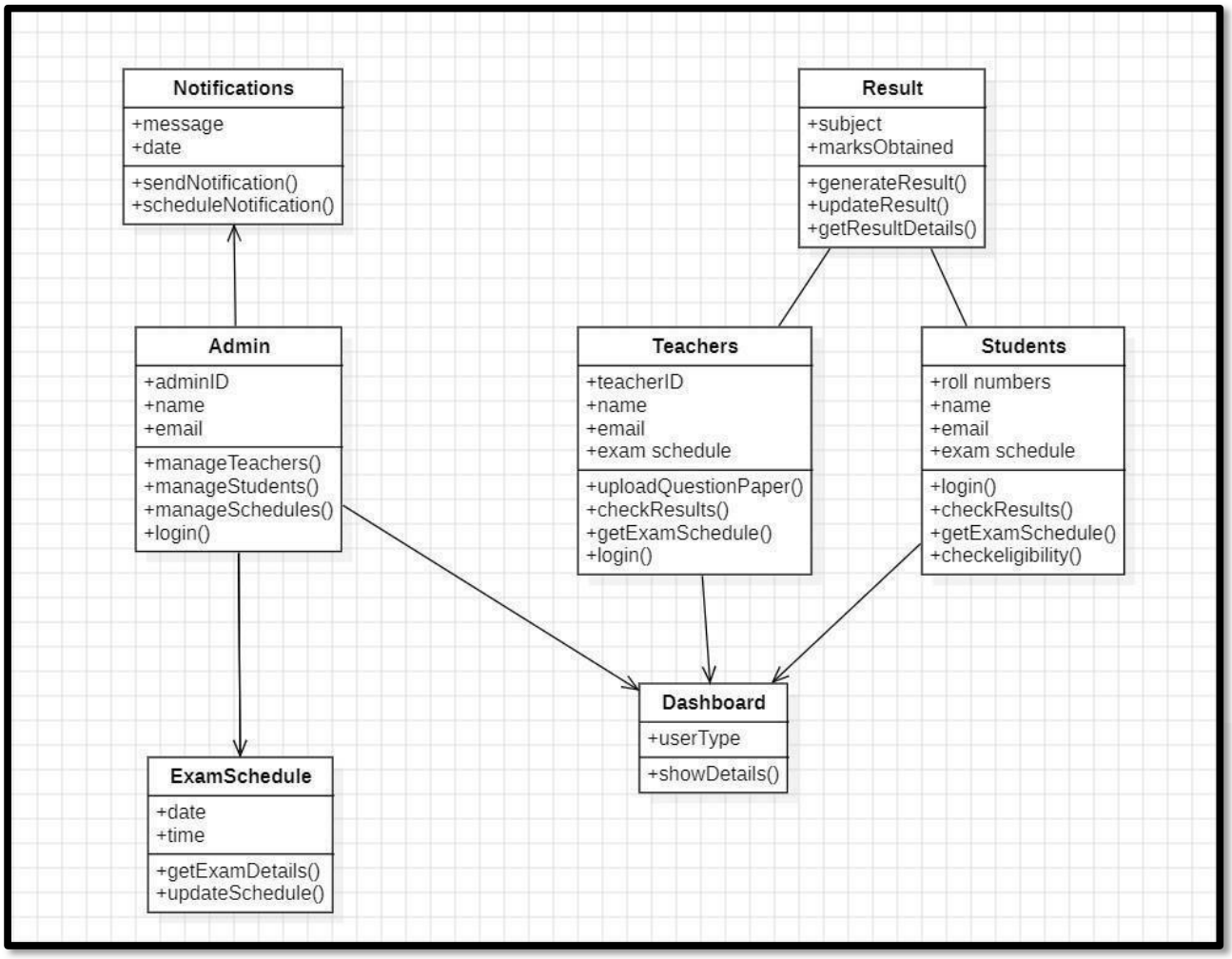
4. Data Flows:

- From Students to Manage Results:
    - Students request results and update their profiles.
- From Teachers to Manage Results and Examinations:
    - Teachers provide exam data and results.

- From Admin to Manage Users and Course and Faculty Data:
    - Admin manages user roles, courses, and faculty information.

2.3 CLASS DIAGRAM:

A class diagram is a static representation of the structure of a system, illustrating the classes, attributes, methods, and the relationships between them. In the context of the Exam Cell Automation System, a class diagram would depict the key entities such as Students, Faculty, Admin, Exam, and Results, along with their attributes and methods. It would also show how these classes are interconnected, providing a clear blueprint of the system's architecture and data structure.



Key Classes:

1. Admin

- Attributes: `adminID`, `name`, `email`
- Methods:
  - `manageTeachers()`: Manages teacher information.

- `manageStudents()`: Manages student information.
- `manageSchedules()`: Manages the exam schedule.
- `login()`: Logs the admin into the system.

2. Teachers

- Attributes: `teacherID`, `name`, `email`, `exam schedule`
- Methods:
  - `uploadQuestionPaper()`: Uploads the exam papers.
  - `checkResults()`: Checks student results.
  - `getExamSchedule()`: Retrieves the exam schedule.
  - `login()`: Logs the teacher into the system.

3. Students

- Attributes: `roll number`, `name`, `email`, `exam schedule`
- Methods:
  - `login()`: Logs the student into the system.
  - `checkResults()`: Views their results.
  - `getExamSchedule()`: Retrieves the exam schedule.
  - `checkEligibility()`: Checks if they are eligible to take the exam.

4. Dashboard

- Attributes: `userType`
- Methods:
  - `showDetails()`: Displays user-specific details on the dashboard.

5. Notifications

- Attributes: `message`, `date`
- Methods:
    - `sendNotification()`: Sends notifications to users.
    - `scheduleNotification()`: Schedules notifications to be sent later.

6. Exam Schedule

- Attributes: `date`, `time`
- Methods:
    - `getExamDetails()`: Retrieves the details of the exam.
    - `updateSchedule()`: Updates the exam schedule.

7. Result

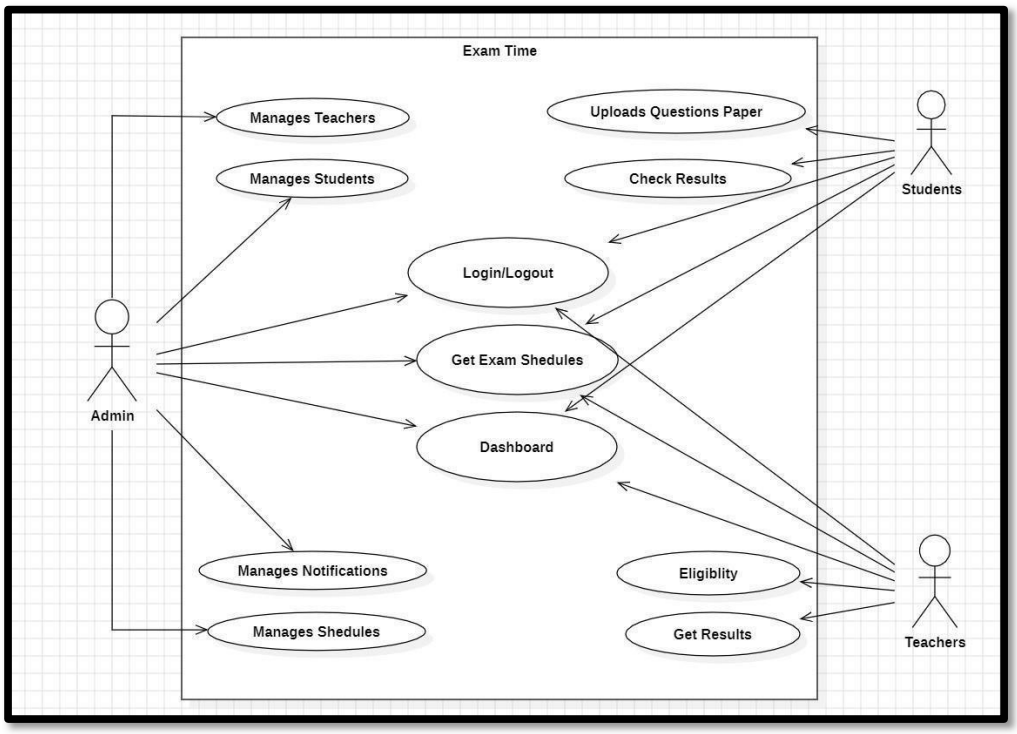
- Attributes: `subject`, `marksObtained`
- Methods:
    - `generateResult()`: Creates results for students.
    - `updateResult()`: Updates existing results.
    - `getResultDetails()`: Retrieves details of the results.

This diagram captures the relationships and functionalities required for managing exams, from scheduling to results, while ensuring effective communication between Admin, Teachers, and Students.

2.4 USE-CASE DIAGRAM:

A use case diagram serves as a visual tool to depict how a system interacts with its users and external entities, providing a clear overview of the system's functionalities and the roles of different actors. For the Exam Cell Automation System, a use case diagram can illustrate the interactions between various users—such as students, faculty, and administrators—and the core processes within the system.

This type of diagram not only helps in understanding the system's flow and user interactions but also aids in identifying the different functionalities the system must support, ensuring all user needs are addressed.



Key Components:

1. **Actors:**

- Admin: Manages teachers, students, notifications, and schedules.
- Teachers: Upload question papers, manage eligibility, and access results.
- Students: Check results, get exam schedules, and access uploaded question papers.

2. **Use Cases:**

- Login/Logout: All users log in or out of the system.

- Dashboard: Provides an overview of the system for users.
- Get Exam Schedules: Allows users to view the exam timetable.
- Manages Teachers/Students: Admins handle teacher and student records.
- Manages Notifications/Schedules: Admins send out important information and set exam dates.
- Uploads Question Papers: Teachers upload the exam questions.
- Check Results: Students view their exam results.
- Eligibility: Teachers check if students are eligible to take exams.
- Get Results: Teachers access exam results.

3. Workflow:

a. Admin Actions:

- Manages teachers and students' information.
- Sets up exam schedules and sends notifications.
- Oversees the general operation of the system.

b. Teacher Actions:

- Uploads exam papers.
- Checks students' eligibility for exams.
- Accesses and manages results after exams.

c. Student Actions:

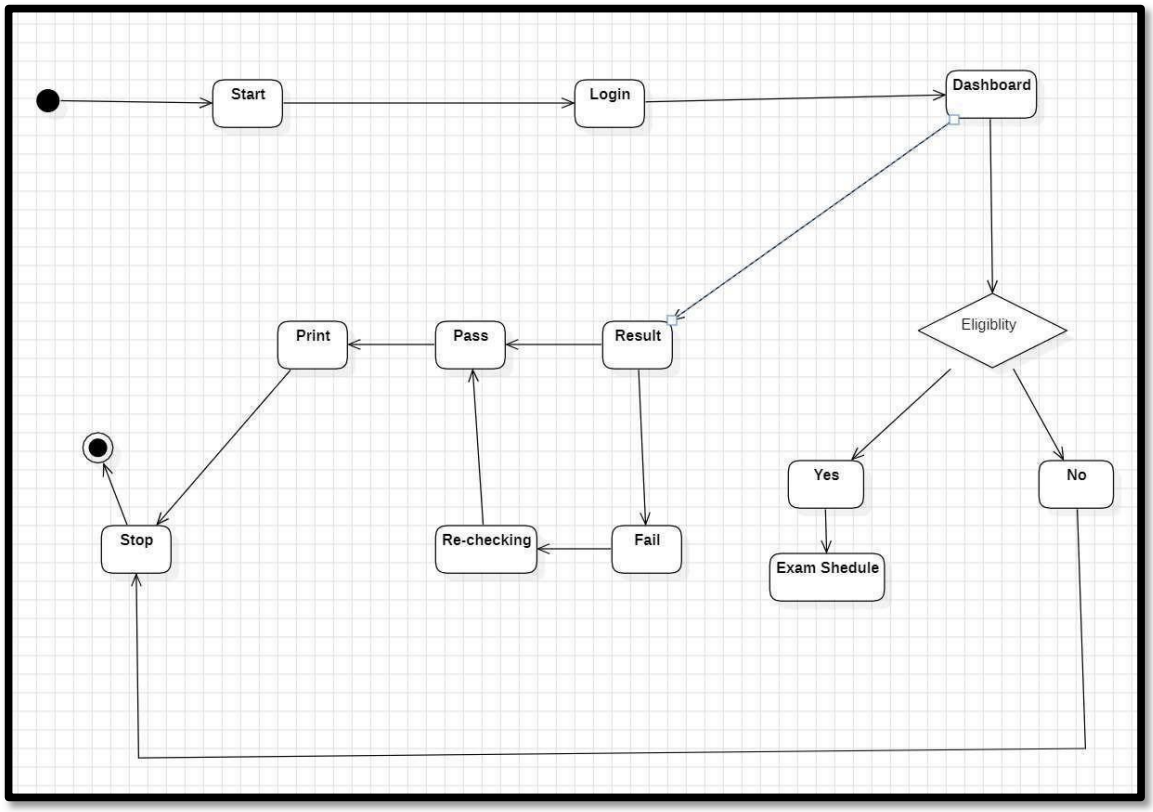
- Logs in to check exam schedules.
- Views their results.
- Accesses uploaded exam papers for reference

In summary, the admin manages the entire process, while Teachers and Students interact with the system based on their specific needs. The system ensures a streamlined exam process from scheduling to result management.

2.5 ACTIVITY DIAGRAM:

Activity Diagrams are used to illustrate the flow of control in a system and refer to the steps involved in the execution of a use case. It is a type of behavioral diagram and we can depict both sequential processing and concurrent processing of activities using an activity diagram that is an activity diagram focuses on the condition of flow and the sequence in which it happens.

1. Student Activity Diagram:

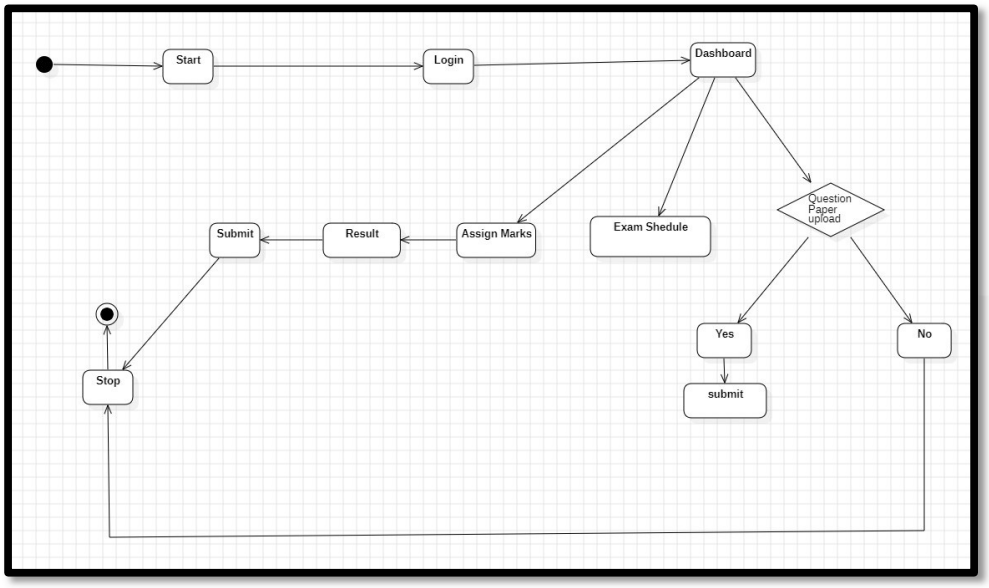


1. The process starts with the student logging into the application.
2. Once logged in, the student will see a dashboard displaying their details, including their name, roll number, and course information etc.
3. Next, student will check whether they match the eligibility criteria to appear for examination or not.
4. If they do meet the criteria, they will be able to view the exam schedule and room assignments. If not, they will not have access to these details.

5. After the exam, if the student passes, they can view their report card. If they fail, they will have the option to request a paper rechecking

6. The process concludes after this.

- **Teacher Activity Diagram**



1. The process begins with teacher logging into the app.

2. Once logged in, the teacher will see a dashboard displaying their details, including their name, email etc.

3. Next, the teacher can upload sets of question papers for their respective subjects on the app.

4. Teachers can receive exam schedule notifications.

5. They will also receive room allotment for invigilation..

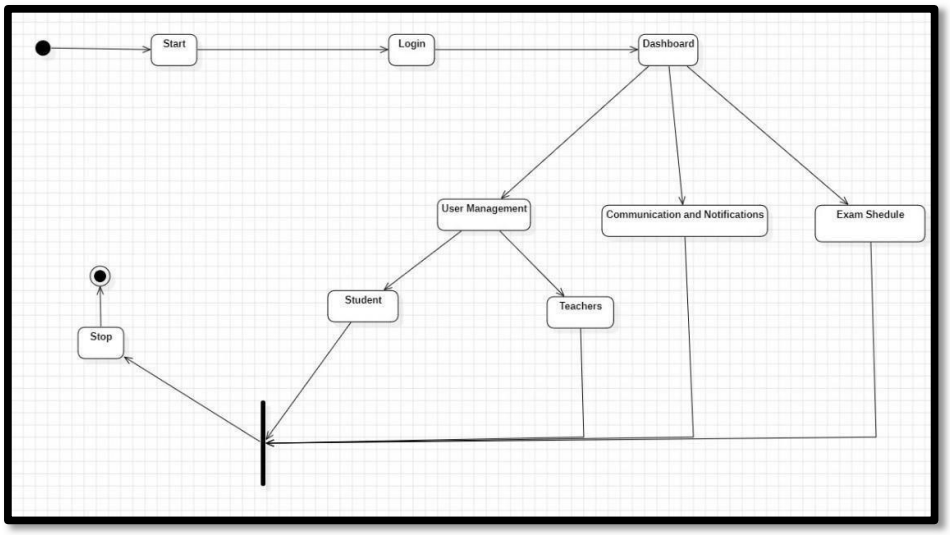
6. Teacher will enter marks to the students, with the help of which the process of student's report card generation will be proceeded.

7. Students who fail and choose to request a recheck will be listed for review.

8. Teachers can recheck their papers online as well.

9. The process would conclude here.

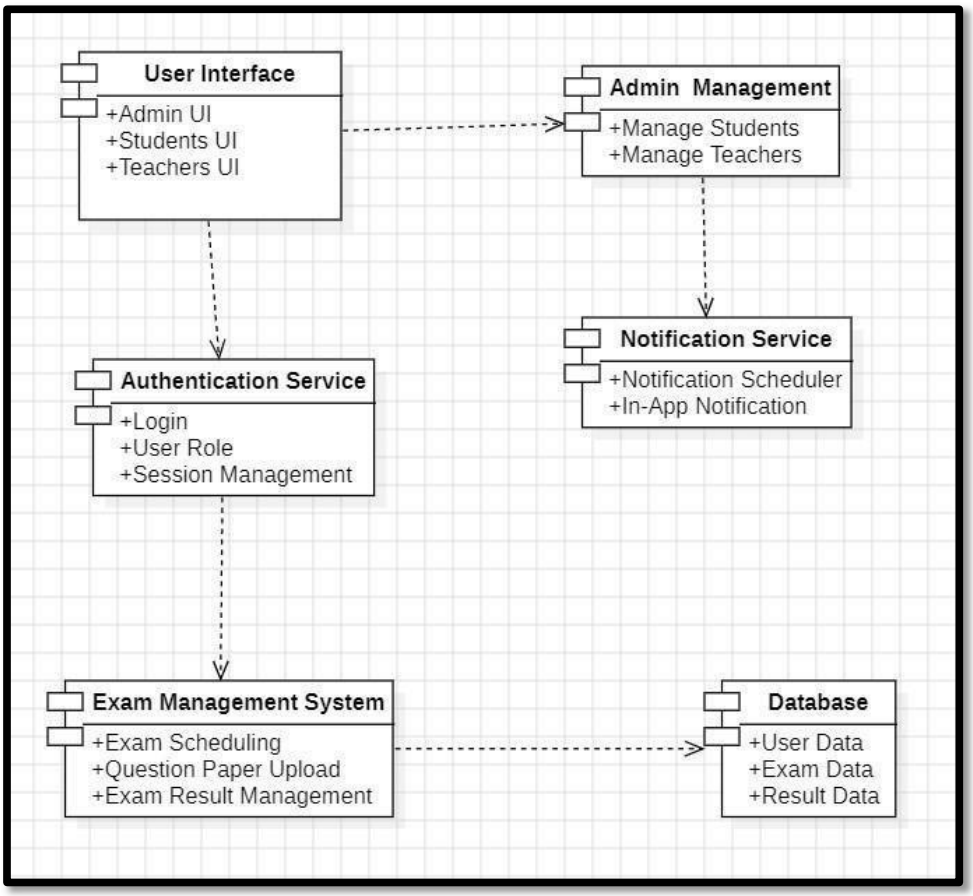
- **Admin Activity Diagram**



1. The process starts with Admin logging in the app.
2. Once logged in, the admin will see a dashboard displaying their details, including their name, email etc.
3. Admin can manage the user (Faculty and Students) ‘s data. Management includes adding, updating or deleting of the data.
4. Admin can send attachments for example, exam schedule.
5. They can also send important notifications such as exam related notices to the users.
6. This will conclude the overall process for the admin.

2.6 COMPONENT DIAGRAM:

A component diagram that shows the structure of a system in terms of its major components and their interactions. For an Exam Cell Automation System, it visually represents key modules like Exam Scheduling, Result Processing, and Portals for students and faculty. This diagram helps in understanding how different parts of the system work together, facilitating better design and implementation.



1. **Key Components:**

    1. **User Interface (UI)**
        - Components:
            - Admin UI
            - Student UI
            - Teacher UI

    2. Authentication Service

        - Responsibilities:

- Manages user login and logout processes.
- Ensures secure access for admins, students, and teachers.

3. Exam Management System

- Components:
  - Exam Scheduling Module.
  - Question Paper Upload Module.
  - Exam Result Management Module.
- Responsibilities:
  - Allows admins and teachers to manage exam schedules.
  - Handles the uploading of question papers.
  - Manages the submission and processing of exam results.

4. User Management System

- Components:
  - Manage Students Module
  - Manage Teachers Module
- Responsibilities:
  - Admins can manage teacher and student data.

5. Notification Service

- Responsibilities:
  - Sends notifications regarding exam schedules, results, and other important updates.

6. Database

- Components:
  - User Data Store
  - Exam Data Store

- Result Data Store

- Responsibilities:

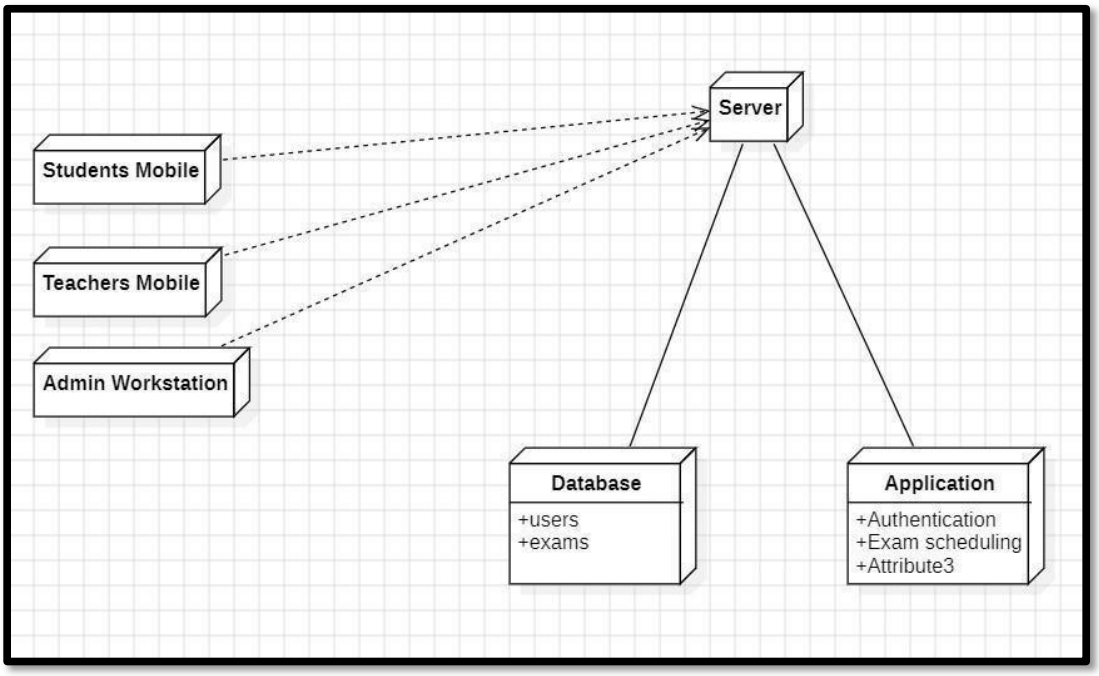
  - Stores and manages all relevant data related to users, exams, and results.

Diagram Description:

- Actors are connected to the Authentication Service and interact with the Exam Management System and User Management System.
- The Exam Management System and User Management System are connected to the Database for storing and retrieving data.
- The Notification Service interacts with the Exam Management System to send out alerts and updates to users.

2.7 DEPLOYMENT DESIGN:

A deployment diagram is a UML diagram that illustrates how software components are distributed across hardware nodes in a system. For an Exam Cell Automation System, it shows how different parts of the system, like the Exam Scheduling and Result Processing modules, are deployed on various servers or devices. This helps in understanding the system's physical architecture and ensures that components are effectively managed and communicated.



- Components

1. Client Devices:

- Students Mobile: Represents mobile devices used by students to interact with the system.
- Teachers Mobile: Represents mobile devices used by teachers.
- Admin Workstation: Represents the workstation used by the admin to manage the system.

These devices connect to the system through a network (most likely the internet), represented by dashed lines leading to the server.

2. Server:

- The central server that acts as a gateway for client devices to access system services. It connects

the client devices (Students Mobile, Teachers Mobile, Admin Workstation) to the backend components (Database and Application).

3. Database:

- Users Table: Contains data related to the users of the system (students, teachers, admins), including personal details, roles, and credentials.
- Exams Table: Stores data related to exams, such as exam schedules, exam papers, results, and eligibility criteria.

The server interacts with this database to perform various operations like retrieving user information, storing exam data, and so on.

4. Application:

- Authentication: Handles user authentication, ensuring that only authorized users (students, teachers, admins) can access the system.

- Exam Scheduling: Manages the scheduling of exams, including creating, updating, and deleting exam schedules.

- Attribute: This is a placeholder for an additional attribute or service. In a real system, this might represent another functional module, such as a Notification Service, Result Processing, or User Management.

- Interactions

1. **Client Devices to Server:** The client devices (students, teachers, admin) interact with the server via the network to request services like login, viewing schedules, uploading exam papers, or checking results.
2. **Server to Database:** The server interacts with the database to store or retrieve information as needed by the requests. For instance, when a student logs in, the server will verify their credentials against the data in the Users Table.
3. **Server to Application:** The server processes requests by using the application logic. For example, it handles authentication, manages exam schedules, and could potentially manage notifications or other functions.

- Overall Functionality

  - The admin can use the Admin Workstation to manage users and exams, making updates that will be stored in the database via the server.

  - Students and Teachers use their mobile devices to interact with the system for tasks such as checking exam schedules, uploading papers, or viewing results.

  - The Server is the central point of interaction between client devices and the backend systems (database and application), ensuring secure and efficient handling of requests.

This architecture represents a typical client-server model where multiple client devices connect to a centralized server, which in turn interacts with a database and application services to fulfill user requests.

CHAPTER 3

SYSTEM PLANNING

Planning is the process of defining and organizing the steps necessary to achieve a specific goal. It is a crucial phase in developing an Exam Cell Automation System. A well-structured and detailed plan is essential for ensuring the effective implementation and success of the system, which includes functionalities like Exam Scheduling, Result Processing, and Portals for students and faculty.

Here are the essential steps to follow during the system planning phase:

1. Defining

- The Exam Cell Automation System is designed to streamline and automate the entire exam process within an educational institution.
- This system will manage the exam schedules, room allocations, eligibility checks, result processing, and communication between students, teachers, and the admin.
- The system will be accessible through an Android app - Exam Time, providing a user-friendly interface for all stakeholders.

2. Setting Clear Objectives

- Automate Exam Scheduling: Allow the admin to schedule exams and allocate rooms efficiently.
- Manage Eligibility: Enable teachers to view student eligibility for exams.
- Question Paper Submission: Facilitate secure and easy submission of question papers by teachers.
- Result Processing: Automate the generation, updating, and distribution of result cards.
- Communication: Provide a platform for sending notifications and updates to students and teachers.
- User Management: Allow the admin to manage users, including adding, updating, and

deleting student and teacher profiles.

- User-Friendly Interface: Ensure the app is intuitive and easy to navigate for students, teachers, and admins.

3. Choosing a Platform

- Development Environment: Android Studio
- Programming Language: Java
- Reasoning: Android Studio is the preferred IDE for Android app development, offering a robust environment with tools and features suited for building, testing, and deploying apps. Java is widely used in Android development, providing extensive libraries and support for building a reliable and efficient application.

4. Data Sources

Institutional Data:

- The app will utilize data provided by the institution, focusing on a specific class. This includes student details (roll numbers, names, contact information), teacher profiles, and class schedules.
- The data will be integrated into the app's backend to support various functionalities like exam scheduling, eligibility checking, and result processing.

5. Technical Indicators

Result Processing:

- The app will use algorithms to automatically generate result cards based on the marks entered by teachers.
- These result cards will include subject-wise marks, overall grades, and percentage calculations. The system will also support rechecking requests and updating results as needed.

6. UI Design

- For Students:

- Login: A secure login screen for students to access their accounts.

- Dashboard: An overview of the student's exam status, upcoming exams, and notifications.

- Exam Eligibility: A feature to check whether the student is eligible to appear for upcoming exams.

- Exam Schedule and Room Allocation: A detailed view of the student's exam timetable and assigned rooms.

- Download Result: After exams, students can download their result cards directly from the app.

- For Teachers:

- Login: A secure login screen for teachers to access their accounts.

- Dashboard: An overview of the teacher’s responsibilities, including scheduled exams and notifications.

- Exam Eligibility Check: A tool to verify if students are eligible to take exams based on attendance and other criteria.

- Exam Schedule and Room Allocation: A view of the exam timetable and room assignments for invigilation.

- Question Paper Submission: A feature for securely submitting exam papers.

- Rechecking: Allows teachers to manage rechecking requests from students and update their marks if necessary.

- For Admin:

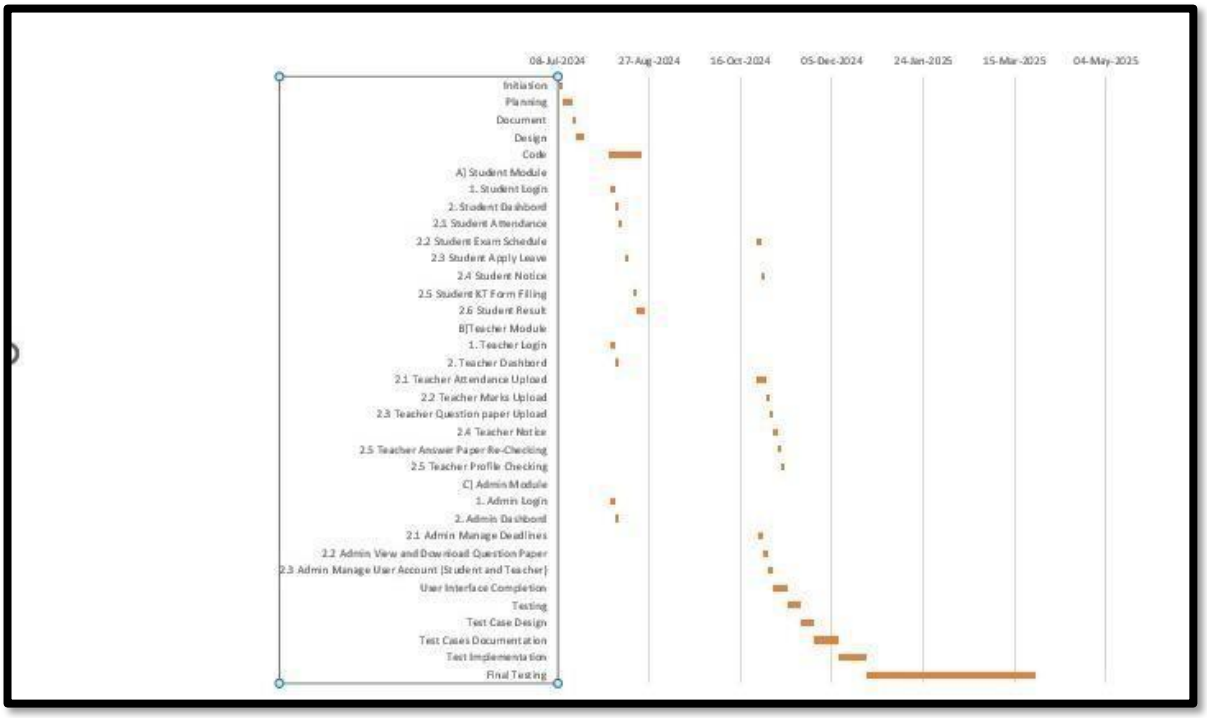
- User Management: An interface to add, update, or delete student and teacher profiles.

- Notifications: A feature to send important updates and announcements to all users.

- Exam Scheduling: Tools to set up exam schedules, allocate rooms, and manage overall exam.

3.1 GANTT CHART:

A Gantt chart is a visual project management tool that displays tasks or activities over time. It uses bars to represent the duration of each task, showing start and end dates along a timeline. This chart helps in tracking progress, identifying dependencies between tasks, and ensuring that the project stays on schedule.



CHAPTER 4

COST ANALYSIS & BENEFIT & EFFORT ESTIMATION

♣ COST ANALYSIS:

Cost analysis is the process of evaluating all the costs involved in a project to determine the total expenditure required. It helps in identifying, estimating, and comparing the expenses associated with different aspects of the project, such as development, infrastructure, marketing, and maintenance. By conducting a thorough cost analysis, you can budget effectively and make informed decisions about resource allocation.

For our Exam Cell Automation app, the cost will depend on several factors, including the complexity of the app. Consider expenses related to development, cloud services, testing, and ongoing maintenance when estimating the total cost.:

1. **Development Time:** The time it takes to develop an app can vary greatly based on its complexity.

2. **Developer's Rate:** The hourly rate of a developer can vary widely depending on their experience and location. As this is our student project so there is no developer rate.

3. **Software and Tools:** Depending on the platform and programming language you choose, there may be software and tool costs involved. The project was developed using free software tools and platforms, meaning no costs were incurred for software or tools during development.

4. **Maintenance and Support::** Once the exam cell automation system is developed and deployed, regular maintenance is required to ensure smooth operation. This includes updating the software, fixing bugs, and ensuring that the system adapts to any new academic policies or exam schedules. Since this is a student project, the maintenance and support are handled in-house without incurring additional cost.

♣ BENEFITS:

Exam cell automation provides significant benefits by streamlining various exam-related processes such as scheduling, result management, attendance tracking, and grading. It minimizes manual work and human error, ensuring consistency, accuracy, and efficiency. Automated systems allow for faster data processing, enabling real-time updates and reducing administrative burdens. The system can be accessed 24/7, offering flexibility to students, faculty, and administrators. By maintaining objectivity and fairness, it eliminates bias in grading and attendance records. Moreover, automated notifications, reporting, and analytics offer improved decision-making and transparency across the exam lifecycle.

Benefits:

1. **Automation of Processes:** Automates tasks like exam scheduling, result generation, attendance management, and grading, reducing manual effort and errors.
2. **Increased Efficiency:** The system processes data quickly and accurately, speeding up time-consuming tasks like result processing and exam setup.
3. **Consistency and Accuracy:** Ensures standardized processes and reduces errors, making sure that exam schedules, grades, and attendance records are accurate and fair.
4. **Objectivity and Fairness:** Removes human bias from grading, scheduling, and attendance tracking, ensuring decisions are based purely on rules and criteria.
5. **Real-time Updates:** Provides real-time information about exam schedules, results, and student performance, improving communication and decision-making.

♣ EFFORT ESTIMATION

The Constructive Cost Model (COCOMO) is a widely used software estimation model that helps predict the effort, cost, and time required to develop a software project. It can be particularly useful for your Exam Cell Automation System to estimate the resources needed for development.

COCOMO Methods:

COCOMO estimates are based on the size of the project, usually measured in lines of code (LOC), and a set of factors that account for the complexity of the project. There are three different COCOMO models:

- Basic COCOMO: Provides a rough estimate based on the size of the project.
- Intermediate COCOMO: Takes into account more factors like the software's complexity, team experience, and the use of tools.
- Detailed COCOMO: Further refines estimates by considering all phases of the software development life cycle.

Step 1: Estimate Lines of Code (LOC)

To apply COCOMO, we first need an approximation of the size of the project in terms of Lines of Code (LOC).

Since LOC estimates can vary significantly based on the project type:

- Student Module: 7 sub-modules (assume 500 LOC per sub-module) → 7 × 500 = 3500 LOC
- Teacher Module: 6 sub-modules (assume 400 LOC per sub-module) → 6 × 400 = 2400 LOC
- Admin Module: 4 sub-modules (assume 600 LOC per sub-module) → 4 × 600 = 2400 LOC
- So, the total estimated LOC for the entire project is:
- 3500+2400+2400=8300 LOC
- Now, convert the total LOC to KLOC (Kilo Lines of Code):

  8300LOC=8.3KLOC

Step 2: Choose the Type of Project

Since this is a college project, it can likely be categorized as an organic project (small, simple, and well understood), so we'll use the constants for an organic project:

a=2.4

b=1.05

Step 3: Calculate Effort (Person-Months)

Using the basic COCOMO formula for effort estimation:

Effort=2.4× (8.3) ^1.05

The estimated effort for the "EXAMTIME" project, based on the COCOMO model, is approximately 22.14 person-months.

Step 4: Determine Duration and Team Allocation

Since 3 people are working on the project, we can estimate the duration of the project as follows:

Duration (in months) = Effort (person-months)/ Team Size = 22.14/3

The estimated duration for the project, with 3 people working, is approximately 7.38 months.

Summary:

Effort: 22.14 person-months

Team size: 3 people

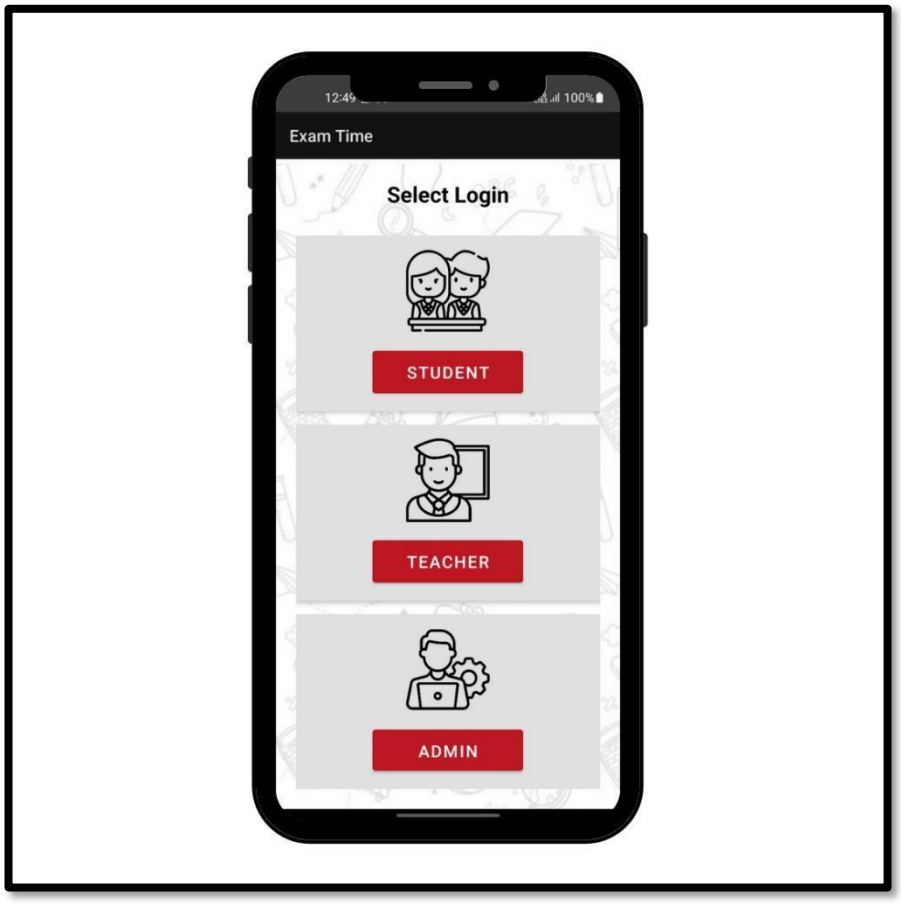
Estimated duration: 7.38 months

This gives you a rough cost estimation for the project based on the COCOMO model.

CHAPTER 5

IMPLEMENTATION AND MODULE WISE EXPLANATION

1. Profile Selection:



1. Login:

- College Email: Enter the email provided by your college, like "devanshi@gmail.com" in the email field.
- Login Button: Click the Login button to proceed.

2. Password:

- College Password: Enter the password assigned by your college. It's masked for privacy.
- Forgot Password: Click this link if you need to reset your password.

2. Login Process:



1. Login:

- College Email: Enter the email provided by your college, like "devanshi@gmail.com", in the email field.
- Login Button: Click the Login button to proceed.

2. Password:

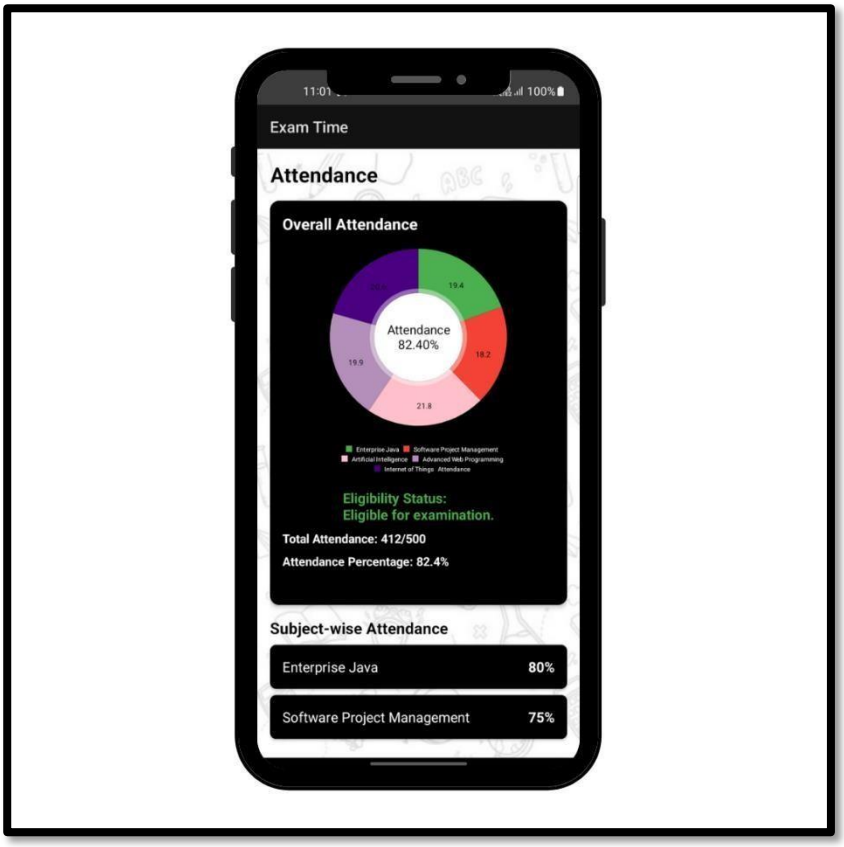
- College Password: Enter the password assigned by your college. It's masked for privacy.
- Forgot Password: Click this link if you need to reset your password..

3. **Student Dashboard :**



1. **Welcome Message:**
    - Student will see a welcome message with their name at the top of the screen.
2. **Notifications:**
    - Important notifications like upcoming exams and deadlines are shown here.
3. **Results:**
    - Access your exam results by clicking the results section.
4. **Apply Leave:**
    - Submit a leave request by clicking the apply leave option.
5. **KT Form Filling:**
    - Fill out the KT (Keep Term) form if applicable by selecting this option.
6. **Profile Settings:**
    - Edit your personal details or change your password under profile.
7. **Profile Settings:**
    - Edit your personal details or change your password under profile.

4. Attendance



1. **Overall Attendance:**

- View your total attendance percentage on this page.
- Check if you are eligible or not eligible for exams based on the overall percentage.
- The eligibility status will be displayed clearly (e.g., "Eligible for exams" or "Not eligible").

2. **Subject-Wise Attendance:**

- Scroll down to see attendance details for each subject (e.g., IoT, AWP, Java, AI).
- Each subject will show Attendance percentage for the specific subject.
- Subjects with low attendance will be marked for attention.

5. Result:



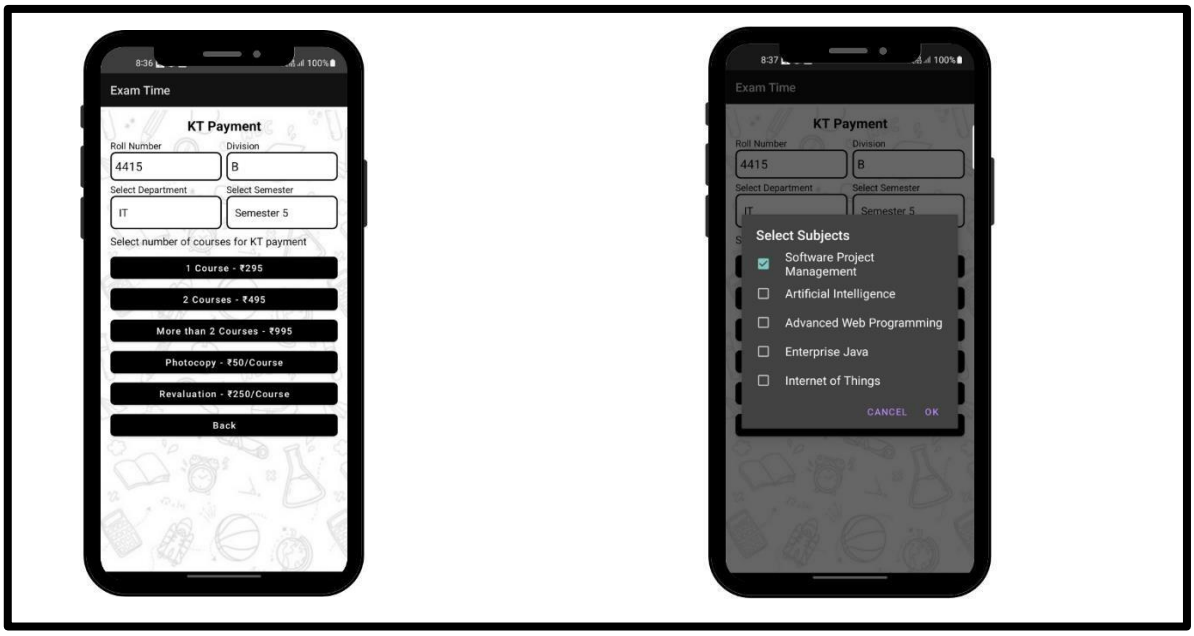
1. **Basic Details:**

- Your name, course, and division will be displayed at the top of the page.
- Marks Breakdown.

2. **For each subject :**

- CE1 (Continuous Evaluation 1) marks.
- CE2 (Continuous Evaluation 2) marks.
- Semester-End exam marks.
- Practicals marks.
- Total Marks and Percentage.
- Below the individual marks, the total marks for each subject will be calculated.
- Student overall percentage for all subjects combined will be shown at the bottom of the page.

6. KT Payment



1. Enter your Roll Number and Division.

2. Select your Department and Semester.

3. Choose the number of courses for KT payment:

- 1 Course – ₹295
- 2 Courses – ₹495
- More than 2 Courses – ₹995

4. Additional options:

- Photocopy – ₹50/Course
- Revaluation – ₹250/Course

5. Click Back if you need to return to the previous screen.

- A list of available subjects will appear.
- Select the subjects for which you need to make a KT payment. For example, check Software Project Management or others like Artificial Intelligence.
- After selecting, click OK to confirm.
- Click Cancel if you want to exit without making a selection.

7. Student Leave



1. **Leave Application Format:**
    - Add a text area with a leave application template (From, To, Subject, Reason).
2. **Date Pickers:**
    - Include "From Date" and "To Date" pickers for selecting dates.
3. **Submit Button:**
    - Add a button to submit the application.

Teacher Modules

1. Teacher Login:



Login:

- College Email: Enter the email provided by your college, like "devanshi@gmail.com", in the email field.
- Login Button: Click the Login button to proceed.

Password:

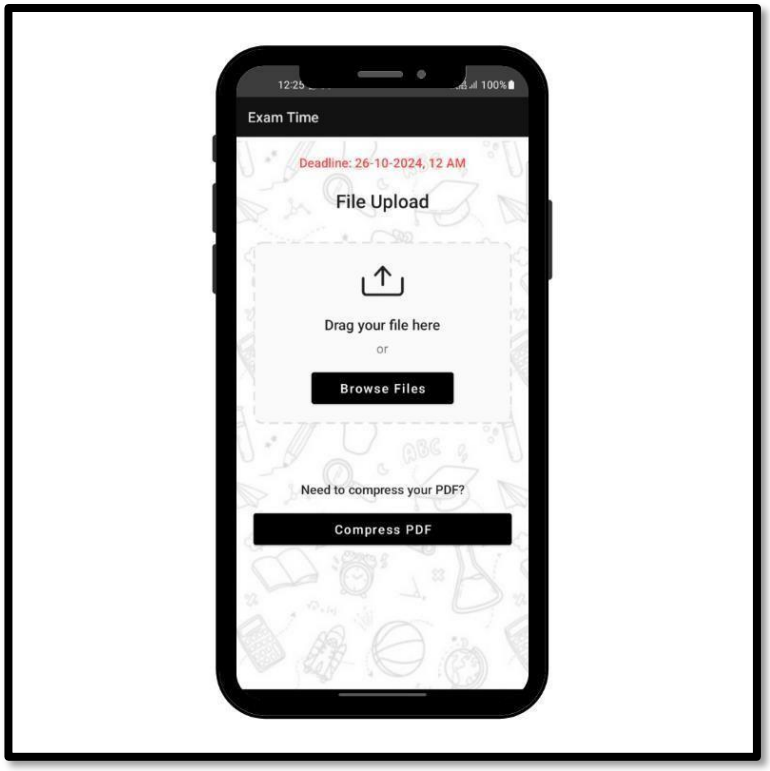
- College Password: Enter the password assigned by your college. It's masked for privacy.
- Forgot Password: Click this link if you need to reset your password.

2. Teacher Dashboard:



1. **Attendance Upload:**
    - Submit or manage student attendance records.
2. **Marks Upload:**
    - Input or update students' marks.
3. **Question Paper Upload:**
    - Upload exam papers for upcoming tests.
4. **Notice:**
    - View or post important notices.
5. **Answer Paper Rechecking:**
    - Request or process answer sheet rechecking.
6. **Profile Settings:**
    - Manage and update your personal details.

4. Question Paper Upload:



1. Deadline:

- Make sure to upload your file before the deadline, shown at the top of the page.

2. File upload:

- Drag your file into the designated area.
- Or click "Browse Files" to manually select the file from your device.

3. Compress PDF (if needed):

- If your file is too large, click "Compress PDF" to reduce its size.

Admin Modules

1. Admin Login



1. **Enter Email:**

    - The email field is pre-filled with "admin@tcsc.com." You would enter your admin email here.

2. **Enter Password:**

    - Type your password in the password field (currently masked for security).

3. **Click "Login" Button:**

    - Press the red "LOGIN" button to proceed**.**

4. **Forgot Password Option:**

    - If you've forgotten the password, click "Forgot password?" for recovery options.

2. Admin Dashboard:



1. Deadline Tracker: Allows admins to manage and track important deadlines related to exams.

2. Question Paper: Provides options for creating, editing, or reviewing question papers.

3. User Administration: Manages user roles, permissions, and other administrative tasks related to users.

4. Notice Board: Displays or posts important announcements and updates for users.

CHAPTER 6

TESTING:

Software testing is a critical process in the development lifecycle of any software application, and the automation of exam cells is no different. It involves systematically executing the software under controlled conditions to verify whether it behaves according to specified requirements and meets the desired outcomes. This aligns with the principles of verification and validation, where verification ensures the correct implementation of software components ("Are we building the system right?"), while validation focuses on confirming that the system fulfills its intended goals and user needs

When testing an exam cell automation system, several crucial components must be thoroughly assessed:

1. **Login Functionality:** Testing the login feature is crucial to ensure secure access to the exam platform. This includes verifying the correct validation of student credentials, ensuring that usernames and passwords are accurately validated, and implementing strong authentication mechanisms to prevent unauthorized access
2. **Forgot Password Functionality:** The "forgot password" feature is essential for secure password recovery. Testing this functionality involves verifying the correct processing of password reset requests, ensuring that users can securely regain access to their accounts and restore full functionality in a timely manner.
3. **Automated Exam Generation:** Automated exam creation is a key function of the system. This includes testing the system’s ability to generate exams based on predefined templates or custom parameters. The process must be tested to ensure that questions are drawn accurately, the right amount of questions are presented, and proper time limits are set for each exam.

A. Student Test Cases

Test Case ID	Description	Input	Expected Output	Actual Output	Status
1	Verify student login with valid credentials.	Valid username & password	Student successfully logs in	Student successfully logs in	Pass
2	Verify student login with invalid credentials	Invalid username/ passord	Error message displayed	Error message displayed	Pass
3	Check if student can view attendance records	Student ID	Attendance records displayed	Attendance records displayed	Pass
4	Check if student can view exam schedule	Course ID	Error: Course not found	Exam schedule displayed	Fail
5	Verify that leader board displays correct rankings	Subject ID	Correct rankings displayed	Rankings missing for some students	Fail
6	Test student ability to view notices	Student ID	Notices displayed	Notices displayed	Pass
7	Validate KT form submission	Form details (subject, reason)	Form submitted successfully	Form submitted successfully	Pass
8	Verify result request submission	Student ID	Request submitted successfully	Request submitted successfully	Pass
9	Verify result download feature	Student ID & Exam ID	Result file downloaded	Error: File not available	Fail

B. Teacher Test Cases

Test Case ID	Description	Input	Expected Output	Actual Output	Status
1	Verify teacher login with valid credentials	Valid username & password	Teacher successfully logs in	Teacher successfully logs in	Pass
2	Verify teacher login with invalid credentials	Invalid username/password	Error message displayed	Error message displayed	Pass
3	Test uploading attendance	Course ID, Student List	Attendance uploaded successfully	Attendance records displayed	Pass
4	Verify marks upload feature	Student ID, Marks Data	Marks uploaded successfully	Exam schedule displayed	Pass
5	Validate question paper upload	Subject ID, Question Paper File	Error: Only PDFs allowed	Question paper uploaded successfully	Fail
6	Check if notice board is accessible	Notice ID	Notices displayed correctly	Notices displayed	Pass
7	Verify answer paper rechecking request	Subject ID, Question Paper File	Request processed successfully	Form submitted successfully	Pass
8	Validate consolidated report generation	Course ID, Exam ID	Request processed successfully	Error: Missing Exam ID	Fail
9	Test unauthorized access to marks upload	No authentication marks data	Error: Unauthorized access	Marks uploaded successfully	Fail

C. Admin Test Cases

Test Case ID	Description	Input	Expected Output	Actual Output	Status
1	Verify admin login with valid credentials	Valid username & password	Admin successfully logs in	Admin successfully logs in	Pass
2	Verify admin login with invalid credentials	Invalid username/password	Error message displayed	Error message displayed	Pass
3	Test adding a new user (student/teacher)	User details (name, role, email)	User added successfully	User added successfully	Pass
4	Verify user deletion feature	Invalid User ID	Error : User not found	User deleted successfully	Fail
5	Validate exam schedule creation	Course ID, Exam Date	Schedule added successfully	Error: Invalid date format	Fail
6	Test notice board update functionality	Notice ID, Content	Notices displayed correctly	Notice updated successfully	Pass
7	Verify faculty assignment report generation	Faculty ID, Course ID	Request processed successfully	Report generated successfully	Pass
8	Validate deadline tracker functionality	Deadline data (event, date)	Deadlines displayed correctly	Error: Missing event name	Fail
9	Test unauthorized access to admin panel	No authentication	Error: Unauthorized access	Admin panel accessed successfully	Fail

CHAPTER 7

CONCULSION:

In reflection, our development journey for the Exam Cell Automation System has been a comprehensive exploration from initial concept to final implementation.

At the outset, we carefully defined our objectives, examined the background of exam management, and clarified user requirements through a detailed feasibility study. This foundational work provided a solid base for our subsequent activities. Moving into the system analysis and design phase, we meticulously developed various diagrams and models, including context diagrams, class diagrams, use case diagrams, and activity diagrams. These tools served as blueprints, guiding our development process and ensuring clarity and efficiency throughout.

Our next focus was on system planning, where we constructed a detailed project timeline using a Gantt chart. This helped us manage project milestones and allocate resources effectively.

During the implementation phase, we transformed our conceptual designs into a functional system. Key functionalities such as exam scheduling, result processing, and user interfaces were meticulously developed through careful coding and development efforts.

Following implementation, we engaged in rigorous testing to ensure the system's reliability, functionality, and security. We employed a range of testing methods to identify and address any issues, ensuring a smooth user experience.

As we conclude this phase, our focus shifts to future enhancements, ensuring continuous improvements to meet the evolving needs of the academic community. With a commitment to innovation, we are poised to make a lasting impact in exam management.

CHAPTER 8

FUTURE WORK:

Our team has set a clear and ambitious goal for the successful implementation of the Exam Cell Automation System. We aim to integrate essential functionalities such as exam scheduling, result processing, and secure user interfaces into the system. This will provide administrators and faculty with valuable tools for managing exams efficiently and effectively. Additionally, we plan to ensure that the system is user-friendly and automated, allowing for streamlined management of exam-related tasks without constant manual intervention. Our objective is to deliver a comprehensive solution that simplifies the complexities of exam management.

Looking ahead, once we achieve this initial milestone and the system gains acceptance, we intend to continuously improve it. We will actively seek feedback from users to incorporate additional features and enhancements based on their needs and preferences. Our ultimate goal is to deploy the system and make it accessible to a wider academic community. As the system's adoption grows, we plan to introduce advanced features and support options to enhance its value and ensure ongoing development. This approach will not only benefit users but also reward our team's dedication and efforts, supporting the continued evolution of the system.

CHAPTER 9

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T.Y.B.Sc.IT Project Attendance Sheet

Project Guide: JYOTSNA ANTHAL

Project Title: EXAM CELL AUTOMATION SYSTEM

Sr. No.	Date	Roll No.	P/ Ab	Project Guide Remark	Status (Complete/ Incomplete)	Sign
1	08-07-24	4414	P	To find out the type of system to be developed	complete	06/07/24
		4415	P			
		4416	P			
2	18-07-24	4414	P	More features to be added for pre-exam work	complete	18/07/24
		4415	P			
		4416	P			
3	14-08-24	4414	P	↳ Question paper submission ↳ Semester end reposrts for each course ↳ Student registeration for exam + admission	complete	14/08/24
		4415	P			
		4416	P			
4	22-08-24	4414	P	Find something which is the problem area in any of the institute	complete	22/08/24
		4415	P			
		4416	P			
5	25-11-24	4414	P	Evaluation of an app	complete	25/11/24
		4415	P			
		4416	P			

Sr. No.	Date	Roll No.	P/ Ab	Project Guide Remark	Status (Complete/ Incomplete)	Sign
6	17-12-24	4414	P	Doubt Solving → leave → grace marks	Complete @3	
		4415	P			
		4416	P			
7	28-02-25	4414	P	Verifying the app features	Complete	28/02/25
		4415	P			
		4416	P			
8	03-03-25	4414	P	Verifying the app	Complete	03/03/25
		4415	P			
		4416	P			
9	11-03-25	4414	P	Verifying documentation	Complete	11/03/25
		4415	P			
		4416	P			